

Finite Elements Analysis in Biomedical Engineering

Spring 2017, 376-1712-00L

A report on

Assignment 2:

Solute concentration diffusion in a cartilage tissue
sample: numerical simulation using finite-difference
formulation

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Problem statement

In this problem, transport of solute through a cartilage tissue sample is to be simulated using a 2D explicit finite-difference formulation, and influence of grid resolution and time intervals on the precision of the results is to be discussed.

Physical setup

The cartilage tissue sample is square shaped ($10\text{ mm} \times 10\text{ mm} \times 3\text{ mm}$) and is inserted in a 2D transport chamber. Since the top and bottom sides of the sample are sealed against glass microscope slides, solute transport across these surfaces is not possible, which allows us to consider this as a purely 2D problem (i.e. gradient of concentration normal to these surfaces is zero, so any gradient, and thus flow, must be tangential).

The remaining sides of the sample are in contact with a large volume well-stirred bath which has a solute concentration of 0 mol/mm^3 , and hence we can assume that solute concentration at these sides (in contact with the bath) will be equal to that of the bath at all times, and that the bath concentration is unaffected by the solute transport into it.

The tissue sample in its center contains a square channel ($2\text{ mm} \times 2\text{ mm}$) through which a fluid with solute concentration of 100 mol/mm^3 is being pumped (beginning from $t = 0\text{ s}$), therefore it can be assumed that solute concentration at the inner boundaries of the sample and in the fluid in the channel are equal at all times.

Given these assumptions, the initial solute concentration and geometry are as in the figure 1. below:

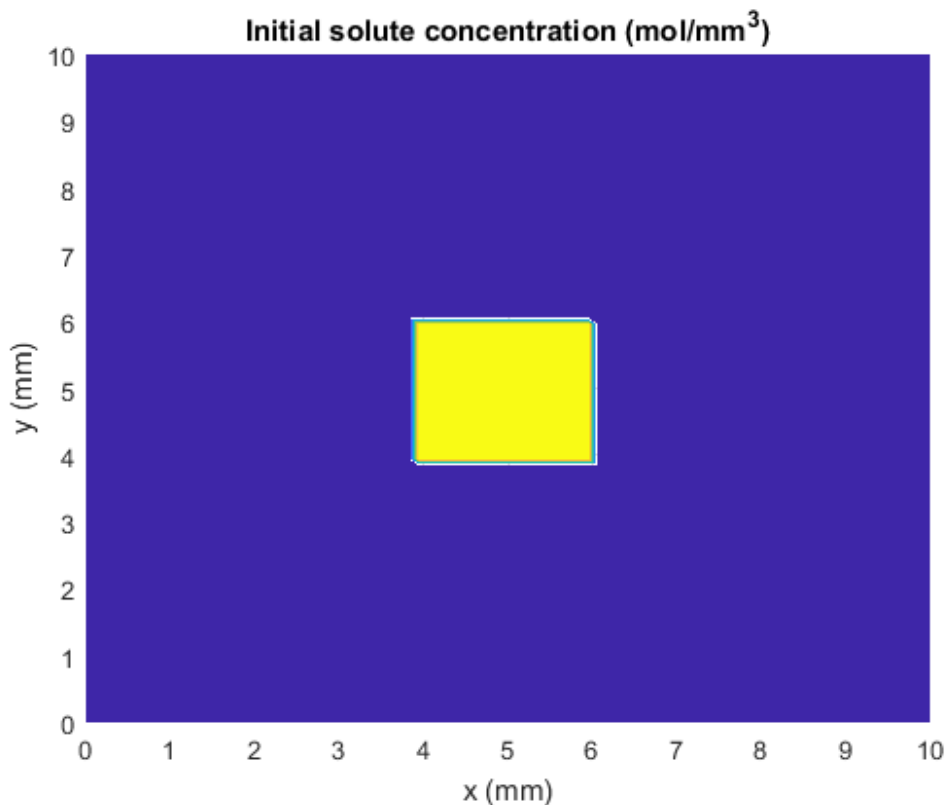


Figure 1: Plot of initial solute concentration and problem geometry. White outline at the sample-channel boundary is of a purely graphical origin due to the way MATLAB® surf() function processes the input data.

The solute diffusivity is anisotropic (but homogeneous), with $D_x = 40 \mu m^2/s$ in the x-direction (horizontal) and $D_y = 20 \mu m^2/s$ in the y-direction (vertical). Due to this, to derive the diffusion equation for anisotropic medium we need to return to Fick's 1st law of diffusion, which written out in full (tensor form) states that the solute flux (ignoring convection) is given by the following:

$$\mathbf{J} = -\mathbf{D} \nabla C = -\begin{pmatrix} D_{xx} & D_{xy} & D_{xz} \\ D_{yx} & D_{yy} & D_{yz} \\ D_{zx} & D_{zy} & D_{zz} \end{pmatrix} \begin{pmatrix} \partial_x C \\ \partial_y C \\ \partial_z C \end{pmatrix}$$

, where c is the solute concentration, $\mathbf{D}$ is the diffusivity tensor and $\mathbf{J}$ is the solute flux. In our case, we are dealing with a 2D geometry and are aligned with the principal axes of the diffusivity tensor, hence the above reduces to:

$$\begin{pmatrix} J_x \\ J_y \end{pmatrix} = -\begin{pmatrix} D_x & 0 \\ 0 & D_y \end{pmatrix} \begin{pmatrix} \partial_x C \\ \partial_y C \end{pmatrix} = -\begin{pmatrix} D_x \partial_x C \\ D_y \partial_y C \end{pmatrix}$$

If we consider an infinitesimally small volume of fluid (aligned with the coordinate axes), only the differences of (normal) solute fluxes at opposite sides cause a change in the solute concentration, from which we have:

$$\frac{\partial C}{\partial t} = -\nabla \cdot \mathbf{J} = -(\partial_x \quad \partial_y) \begin{pmatrix} J_x \\ J_y \end{pmatrix} = -\partial_x J_x - \partial_y J_y = D_x \frac{\partial^2 C}{\partial x^2} + D_y \frac{\partial^2 C}{\partial y^2}$$

, which is the equation we will numerically solve in this problem.

Finite difference formulation

In general, solving any problem using a computer necessitates creating a finite (countable) set of algebraic equations to be solved. One method is to divide – or discretise – all of the problem domains (space, time) into countably many points and to discretise the governing equations as well. This is in principle the finite difference formulation, where all partial derivatives are approximated as finite differences in space and time „grids“.

In this case we divide the spatial domain into a regular grid with even nodal spacing of ΔL in both directions and use second order finite difference approximation for all spatial derivatives. The temporal domain is divided into finite timesteps of length Δt and we use first order finite difference approximation (i.e. as given in problem text). Due to this mismatch, it is expected that reducing the timestep interval will have a stronger effect on the overall precision of the numerical solution.

Denoting with $C_{i,j}^k = C(i, j, k)$ the value of solute concentration at gridpoint (i, j) at time step k and using an explicit formulation, we obtain the following discretised formulation of diffusion equation:

$$\frac{C_{i,j}^{k+1} - C_{i,j}^k}{\Delta t} = \frac{D_x}{\Delta x^2} (C_{i+1,j}^k - C_{i,j}^k + C_{i-1,j}^k) + \frac{D_y}{\Delta y^2} (C_{i,j+1}^k - C_{i,j}^k + C_{i,j-1}^k)$$

, which leads to the final equation we shall implement in the code:

$$C_{i,j}^{k+1} = (1 - 2b - 2ab) C_{i,j}^k + b (C_{i+1,j}^k + C_{i-1,j}^k) + ab (C_{i,j+1}^k + C_{i,j-1}^k)$$

, where we have defined the anisotropy factor as $a = D_y/D_x = 0.5$ and the stability factor as:

$$b = D_x \Delta t / \Delta x^2$$

Iterative update, Initial and Boundary conditions

To obtain the solute concentration at the next timestep at a certain gridpoint, values of solute concentrations at the current timestep from the same and the neighboring gridpoints are combined together as in the discrete diffusivity equation (5 gridpoints are used, hence the name „5-point stencil“). This process is repeated for all gridpoints and then iteratively repeated for all future timesteps, which gives us the desired result, the time evolution of solute concentration.

However, to run the iterative update we first need to define the initial conditions and the boundary conditions (which hold at all timesteps). Assuming that the initial solute concentration in the sample is zero, we can initially assign this value to the whole sample, while value of 100 mol/mm^3 is assigned to the channel. As discussed previously, the boundary values to be used are 0 mol/mm^3 for the outer boundaries of the sample and 100 mol/mm^3 for the inner boundary of the sample. Since in our implementation the whole channel is included in the computational domain (in order to keep the code simple), diffusion would happen in the channel as well which is not correct, and this is prevented by „reseting“ the solute concentration of gridpoints in the channel at each timestep.

Structure of the code

Given all of the above, the problem of solute transport by diffusion can be numerically solved, and the code (written in MATLAB® R2017a) was structured as follows, with the whole code grouped in the main file named „FEAinBME2017_assignment2_SolverPepic“:

1. Declaring physical parameters (spatial and temporal domain, diffusivities)
2. Declaring simulation parameters (gridpoints, time steps)
3. Assessing if stability criterion is satisfied
4. Declaring initial and boundary conditions
5. Iterative calculation of time evolution of solute concentration (explicit FDM)
6. Plotting the results

Results

In an explicit formulation used here, it is theoretically expected that numerical errors will amplify and produce a „runaway“ effect if the multiplicative factor in front of $C_{i,j}^k$ (in iterative update equation) is smaller than one, which gives us a constraint of: $b \leq \frac{0.5}{1+a} = \frac{1}{3} = 0.33\bar{3}$. When running the simulation it was found that (huge) runaway errors are not visible when $b = 0.3334$ even after 1000 hours of simulation time, but they are already noticeable after 48 hours of simulation time when $b = 0.334$, which closely matches the theoretical prediction.

As a sidenote: since diffusivity is anisotropic it is expected to see a stronger diffusion e.g. along the x-direction than in the y-direction and this causes isoconcentration lines to be ellipses, since on average we have $x_{diff}^2 = D_x t$ and $y_{diff}^2 = D_y t$, giving $\frac{x^2}{D_x} + \frac{y^2}{D_y} = 2t = \text{const.}$ at a certain timestep.

The solute concentrations in the sample were plotted at $t = [0 \ 1 \ 6 \ 24 \ 48] \text{ hours}$ for two different grid resolution (50 and 100 gridpoints) and two different time intervals (3000 and 10000 timesteps) and are shown below (organized in separate columns). Additionally, a plots along the x-direction for $y = 7 \text{ mm}$ at time $t = 48 \text{ hours}$ is provided to aid in assessing precision of the obtained solution.

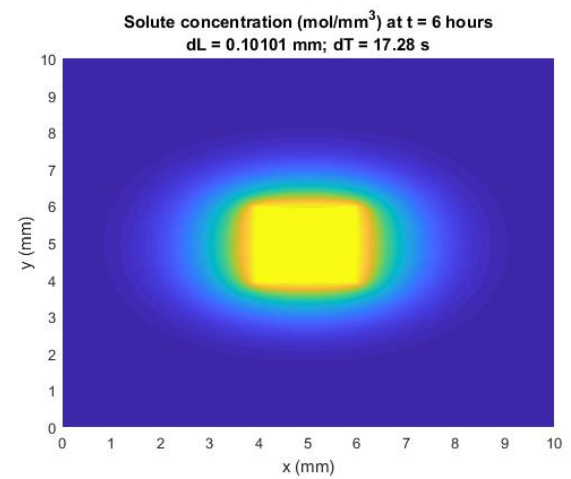
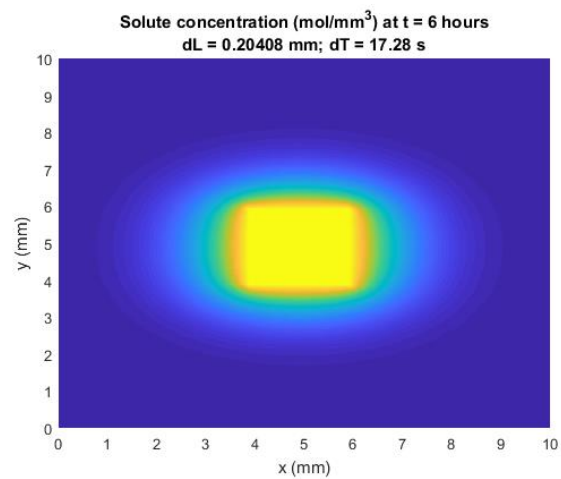
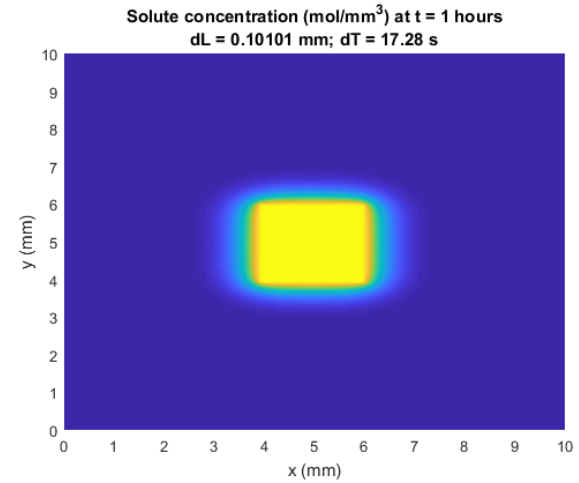
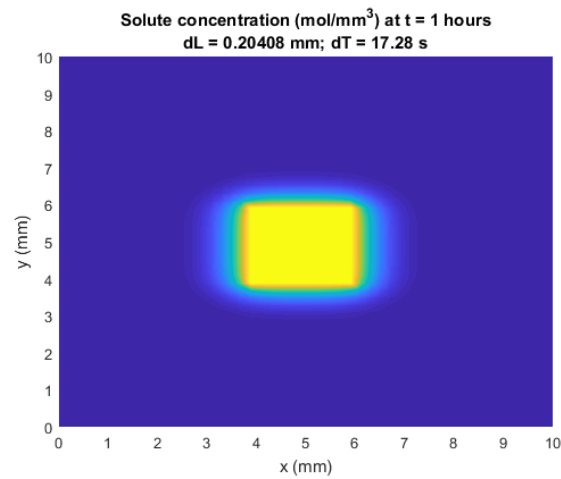
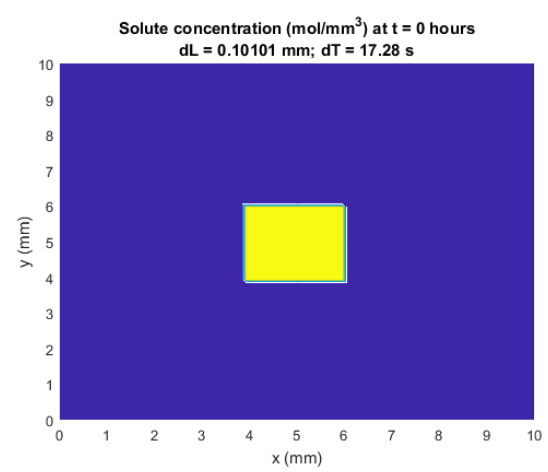
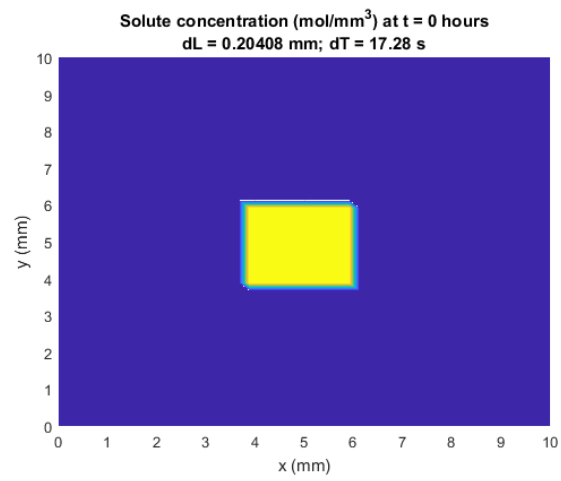
Influence of grid resolution on precision

50 gridpoints

10000 timesteps

100 gridpoints

10000 timesteps



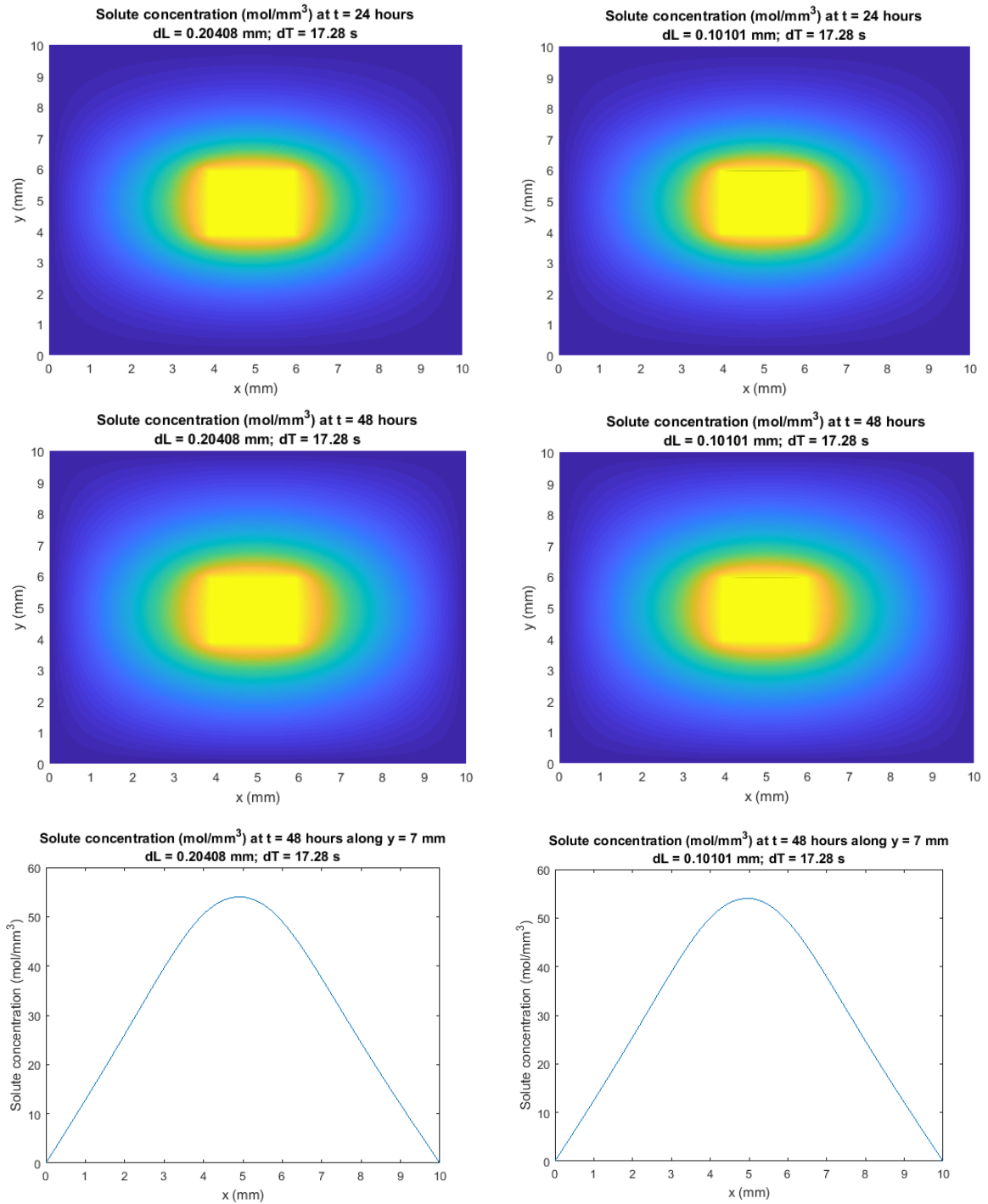
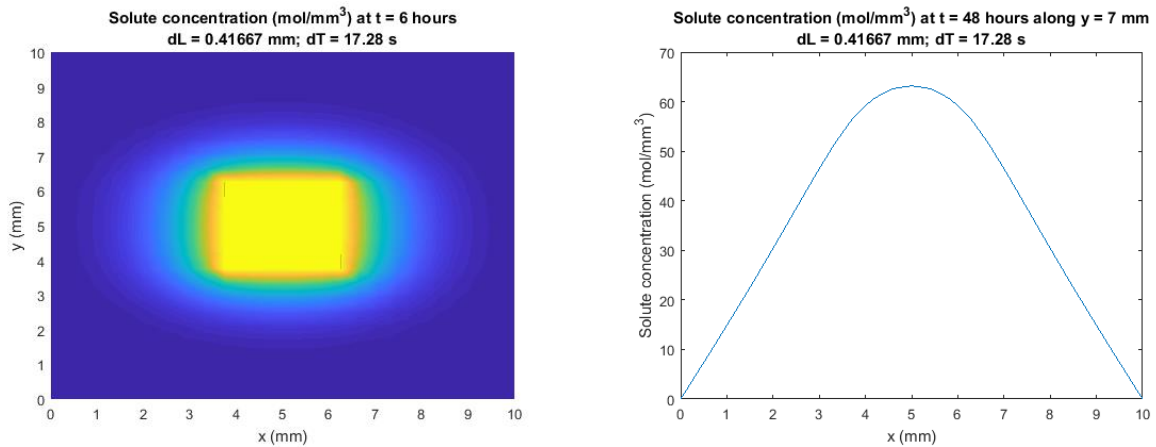


Figure 2: Plots of temporal evolution of solute concentration for two different grid resolutions (50 and 100 gridpoints), with temporal resolution kept identical (10000 timesteps). No obvious differences for chosen grid resolutions can be noticed in these plots.

Excluding the slight errors in graphical rendering (same as in figure 1), both sets of plots seem very similar i.e. there are no obvious differences. Upon closer inspection of the isoconcentration lines in color plots, it seems as that diffusion is slightly faster for lower spatial resolutions (also for 25 gridpoints). Likewise, additional plots showing the concentration plot along $y = 7 \text{ mm}$ do not show any noticeable difference for chosen resolutions. However, a plot with 25 gridpoints does show some difference (shown in figure 3), but for such a coarse mesh this is expected to happen and is of limited practical relevance.



Figures 3: Plot of solute concentration at t = 6 hours for grid resolution of 25 gridpoints and temporal resolution of 10000 timesteps, for the whole domain (left) and along the horizontal line at height y = 7 mm (right). When compared to the same plot for higher grid resolutions, it can be seen that isoconcentration lines have advanced further, implying a (numerical) faster diffusion. A corresponding increase in concentration is shown in the figure right (note the values on the vertical axis). However, for such a coarse mesh no definitive conclusions are to be made.

To investigate the precision further, results from simulations with 25, 50 and 100 gridpoints were saved, after which the solute concentrations at corresponding gridpoints were compared (i.e. each second gridpoint in „50 gridpoint“ data set was compared with each gridpoint in „25 gridpoint data set). Taking the „100 gridpoints“ data set as the one containing the „true“ values, a sum squared difference relative to the data sets can be computed, and from that the relative (sum) error can be obtained (here defined as ratio of SSD and the mean value of concentration in the whole sample). In this case, values of 0.210 and 0.074 were obtained for relative errors for grid resolution of 25 and 50 gridpoints, respectively.

Influence of temporal resolution on precision

Similarly as for the grid resolution, results obtained when changing the temporal resolution (figure 4.) did not show any noticeable differences - including the profile plots at y = 7 mm - even though a stronger effect on precision was expected given that a first order approximation was used for the temporal derivatives, as compared to a second order approximation used for spatial derivatives.

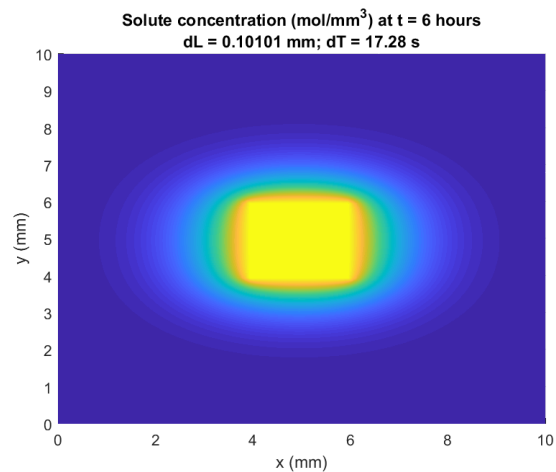
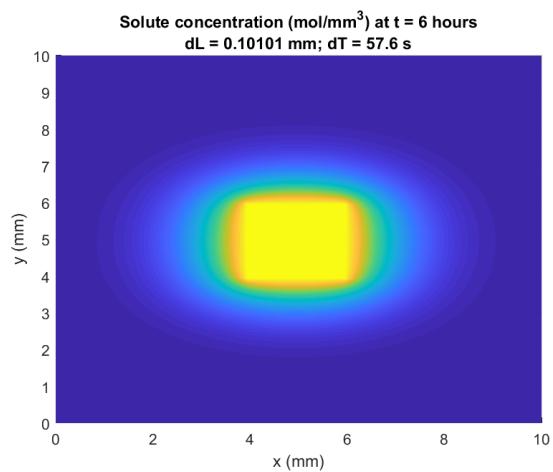
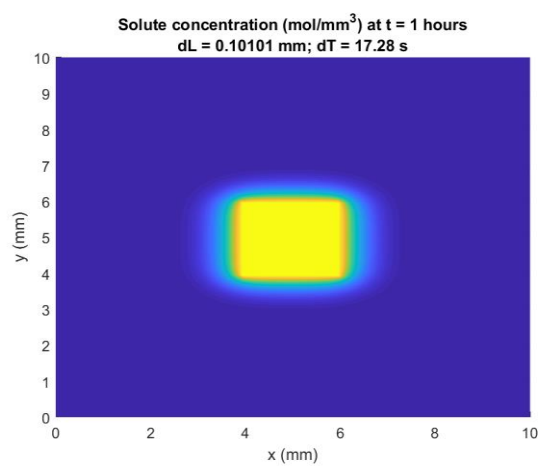
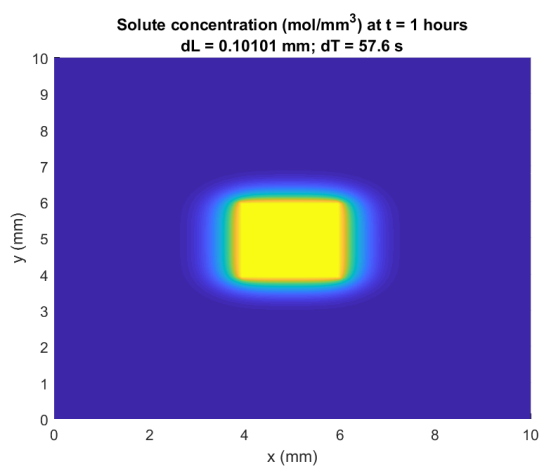
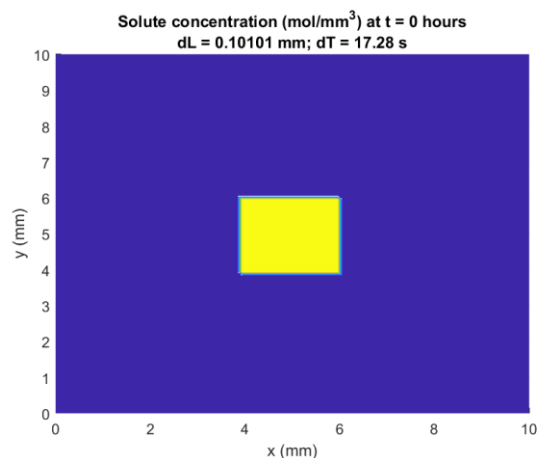
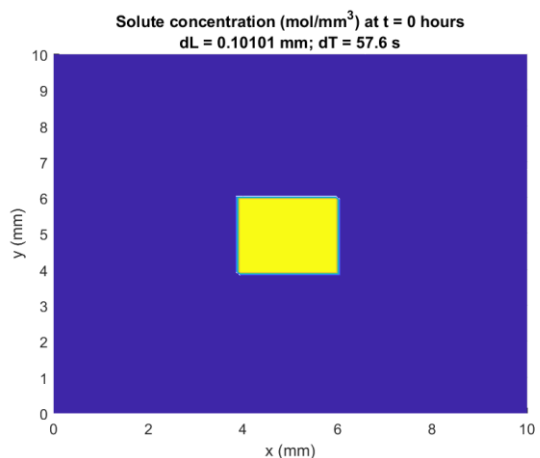
Ideally, results should be obtained for lower resolutions to notice the influence on precision, but these are not possible since stability criterion would be violated. For example, even at only 25 gridpoints it is necessary to have approximately 120 timesteps or more to stay in the stability region, and for 100 gridpoints – and much better choice since we could assume no errors come from spatial discretisation – we need at least ca. 2400 timesteps.

100 gridpoints

3000 timesteps

100 gridpoints

10000 timesteps



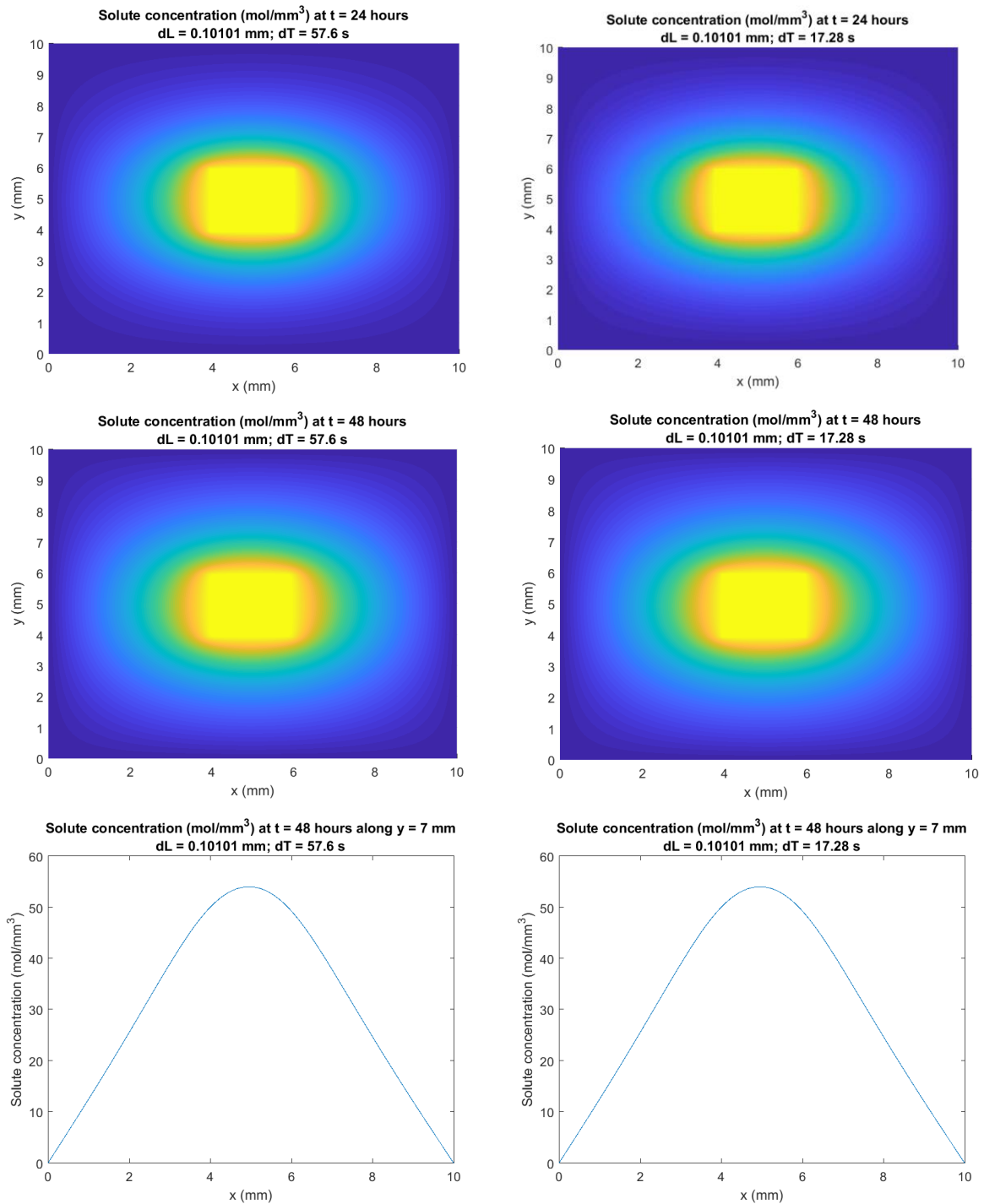


Figure 4: Plots of temporal evolution of solute concentration for two different temporal resolutions (3000 and 10000 timesteps), with grid resolution kept identical (100 gridpoints). No obvious differences for chosen temporal resolutions can be noticed in these plots.

Literature:

1. Benedikt Helgason, Stephen J. Ferguson; Finite Elements Analysis in Biomedical Engineering course; Spring 2017, ETH Zurich
2. MATLAB Documentation - MathWorks® Switzerland